



Advanced Time Series Analysis and Forecasting

Seminar 7: Regime-switching models



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ **AI:** Artificial Intelligence
- ▣ **AIC:** Akaike Information Criterion
- ▣ **AR:** AutoRegressive (model); in event studies: Abnormal Return
- ▣ **BET:** Bucharest Exchange Trading (index)
- ▣ **BIC:** Bayesian Information Criterion
- ▣ **BNR:** Banca Națională a României (the National Bank of Romania)
- ▣ **EM:** Expectation–Maximisation (algorithm)
- ▣ **EODHD:** EOD Historical Data (market-data provider)
- ▣ **FRED:** Federal Reserve Economic Data
- ▣ **GARCH:** Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GDP:** Gross Domestic Product
- ▣ **GNP:** Gross National Product
- ▣ **GPH:** Geweke–Porter–Hudak (estimator)
- ▣ **HICP:** Harmonised Index of Consumer Prices
- ▣ **HMP:** Haas–Mittnik–Paoletta (MS-GARCH, 2004)

Acronyms used in this material (2/2)

- ▣ **HQ**: Hannan–Quinn (information criterion)
- ▣ **LR**: Likelihood Ratio (test)
- ▣ **MCMC**: Markov Chain Monte Carlo
- ▣ **ML**: Machine Learning
- ▣ **MS**: Markov Switching
- ▣ **MS-GARCH**: Markov-Switching GARCH
- ▣ **MS-VAR**: Markov-Switching Vector AutoRegression
- ▣ **MSIAH**: Markov-Switching Intercept, Autoregressive parameters and Heteroskedasticity
- ▣ **MSIAH-AR**: Markov-Switching Intercept, AR parameters and Heteroskedasticity AutoRegression
- ▣ **MSIH**: Markov-Switching Intercept and Heteroskedasticity
- ▣ **MSM**: Markov-Switching Mean (Krolzig notation)
- ▣ **MSM-AR**: Markov-Switching Mean AutoRegression
- ▣ **NBER**: National Bureau of Economic Research
- ▣ **QPS**: Quadratic Probability Score
- ▣ **TVTP**: Time-Varying Transition Probabilities
- ▣ **US**: United States

Today's question and route

- **Question:** how do we infer an unobserved regime, estimate a switching model, decide how many regimes there are, and read the result?
 - ▶ this seminar comes **before** Lecture 7: the section “Prerequisites for Today” gives every definition the tasks use
- Route
 - ▶ Part A: the Hamilton filter and the Kim smoother by hand, one EM step, regime forecasts, information criteria, identification, MS-GARCH variances, Gibbs conditionals
 - ▶ Part B: numpy against statsmodels; Hamilton (1989); how many regimes in Romanian GDP; Romanian inflation; S&P 500 and BET; MS-GARCH for the BET
 - ▶ Part C: EUR/RON in 2025, a new regime or a break?; an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

Exercise map

Task	Question	Type	Model
A1, A2	the Hamilton filter by hand; the Kim smoother and joint probabilities	Solved, Proposed	A1
A3, A4	one EM step; regime forecasts and persistence	Solved, Proposed	A1, A3
A5, A6	information criteria and a bootstrap p -value; identification and the expanded state	Solved, Proposed	A5
A7, A8	MS-GARCH variances (Haas–Mittnik–Paoletta, Gray); Gibbs conditionals	Solved, Proposed	A7, A3
B1, B2	numpy against statsmodels (US GDP); Hamilton (1989)	Solved, Proposed	B1
B3, B4	how many regimes in Romanian GDP growth?; Romanian inflation regimes	Solved, Proposed	B3
B5, B6	bull and bear regimes of the S&P 500 and the BET; MS-GARCH for the BET	Solved, Proposed	B5
C1, C2	EUR/RON in 2025; what is wrong in an AI answer?	Proposed	B5, A1–A8

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
US real GDP, NBER recession quarters	FRED (GDPC1, USRECQ)	quarterly	1947–2026
Hamilton's real GNP data	statsmodels	quarterly	1951–1984
Romanian real GDP; HICP	Eurostat	quarterly; monthly	1995–2026
S&P 500, BET	EODHD	daily, weekly	2000–2026
EUR/RON	BNR reference rate	daily	2023–2026

- In the notebook: `hamilton_filter`, `kim_smoother`, `em_msr`, `best_em`, `msr_ml`, `msm_fit`, `lr_bootstrap`, `msgarch_fit`, `gibbs_ms`; no account or key is needed

Prerequisites for Today (1/7): the switching regression

- MS(K) regression: the coefficients and the variance depend on a hidden regime

$$y_t = x_t' \beta_{S_t} + \sigma_{S_t} \varepsilon_t, \quad \varepsilon_t \sim N(0, 1)$$

- ▶ $S_t \in \{1, \dots, K\}$: the regime at t , not observed; x_t : regressors (constant, lags); β_j, σ_j : coefficients and shock s.d. in regime j
- ▶ S_t is a Markov chain: $p_{ij} = \Pr(S_t = j \mid S_{t-1} = i)$, collected in the transition matrix P (rows sum to 1)
- Special forms [Krolzig \(1997\)](#)
 - ▶ MSIH: intercept and variance switch
 - ▶ MSM (Hamilton 1989): the mean μ_{S_t} switches, $y_t - \mu_{S_t} = \sum_k \phi_k (y_{t-k} - \mu_{S_{t-k}}) + \sigma \varepsilon_t$; ϕ_k : common AR coefficients

Prerequisites for Today (2/7): the Markov chain

▣ Two regimes

- ▶ ergodic probability (long-run share of time in regime 1): $\pi_1 = \frac{1 - p_{22}}{2 - p_{11} - p_{22}}$
- ▶ expected duration of a visit to regime i : $1/(1 - p_{ii})$
- ▶ persistence $\lambda = p_{11} + p_{22} - 1$: close to 1 means long, sticky regimes

▣ Regime forecasts converge to π_1 at the rate λ^h

- ▶ $\Pr(S_{t+h} = 1 \mid S_t = 1) = \pi_1 + \lambda^h(1 - \pi_1)$, $\Pr(S_{t+h} = 1 \mid S_t = 2) = \pi_1 - \lambda^h\pi_1$

▣ Regimes are named by an ordering ($\mu_1 < \mu_2$ or $\sigma_1 < \sigma_2$): the likelihood does not change if labels are swapped

Prerequisites for Today (3/7): the Hamilton filter

□ Notation

- ▶ $Y_t = (y_1, \dots, y_t)$; $\hat{\xi}_{t|s}$: the vector of probabilities $\Pr(S_t = j \mid Y_s)$, $j = 1, \dots, K$
- ▶ $\eta_{jt} = f(y_t \mid S_t = j, Y_{t-1})$: the density of y_t in regime j ; $\odot$: element-wise product

□ **Hamilton filter** [Hamilton \(1989\)](#): predict with the chain, then update with Bayes

- ▶ $\hat{\xi}_{t|t-1} = P' \hat{\xi}_{t-1|t-1}$; $\hat{\xi}_{t|t} = \hat{\xi}_{t|t-1} \odot \eta_t / f(y_t \mid Y_{t-1})$
- ▶ $f(y_t \mid Y_{t-1}) = \sum_j \hat{\xi}_{j,t|t-1} \eta_{jt}$; $\ln L = \sum_t \ln f(y_t \mid Y_{t-1})$; start $\hat{\xi}_{1|0} = \pi$ (ergodic)

Prerequisites for Today (4/7): smoothing and forecasting

- **Kim smoother** Kim (1994): $\Pr(S_t = i, S_{t+1} = j \mid Y_T) = \hat{\xi}_{i,t|t} p_{ij} \hat{\xi}_{j,t+1|T} / \hat{\xi}_{j,t+1|t}$; sum over j for $\hat{\xi}_{i,t|T}$
 - ▶ runs backwards from $\hat{\xi}_{T|T}$; T : the sample length
 - ▶ filtered = what was known at t ; smoothed = with hindsight; never evaluate real-time dating with smoothed probabilities
- Forecast: $\hat{\xi}_{T+h|T} = (P')^h \hat{\xi}_{T|T}$
 - ▶ the predictive density is a mixture $\sum_j \hat{\xi}_{j,T+1|T} N(x' \beta_j, \sigma_j^2)$: a weighted sum of the regime densities

Prerequisites for Today (5/7): EM and testing

- **EM Hamilton (1990):** E-step = Kim smoother; M-step:
$$p_{ij} = \sum_t \Pr(S_{t-1} = i, S_t = j \mid Y_T) / \sum_t \Pr(S_{t-1} = i \mid Y_T)$$
 - ▶ p_{ij} = expected number of $i \rightarrow j$ transitions over expected visits to i
 - ▶ β_j, σ_j^2 : least squares weighted by $\hat{\xi}_{j,t|T}$; EM finds local maxima: use many starting values, then numerical ML
- **Testing $K = 1$ against $K = 2$:** LR is not χ^2 Hansen (1992), Garcia (1998)
 - ▶ under H_0 , p_{11} and p_{22} are not identified, the null is on the boundary and the score is zero
 - ▶ parametric bootstrap: simulate from the fitted null, re-estimate both models,
 $p = (1 + \#\{LR^* \geq LR\}) / (B + 1)$; LR^* : the B bootstrap statistics

Prerequisites for Today (6/7): information criteria and MS-GARCH

- ▣ AIC = $-2 \ln L + 2k$, BIC = $-2 \ln L + k \ln T$, HQ = $-2 \ln L + 2k \ln \ln T$: the smallest value wins
 - ▶ k : the number of free parameters, including the $K(K-1)$ free transition probabilities
- ▣ **MS-GARCH**: the naive $h_t = \omega_{S_t} + \alpha_{S_t} \varepsilon_{t-1}^2 + \beta_{S_t} h_{t-1}$ depends on the whole regime path (K^t histories)
 - ▶ h_t : conditional variance; ε_{t-1} : last shock; $\omega_j, \alpha_j, \beta_j$: GARCH parameters of regime j
 - ▶ Haas, Mittnik and Paolella (2004): $h_{j,t} = \omega_j + \alpha_j \varepsilon_{t-1}^2 + \beta_j h_{j,t-1}$, K GARCH processes in parallel
 - ▶ Gray (1996): $h_{j,t} = \omega_j + \alpha_j \varepsilon_{t-1}^2 + \beta_j h_{t-1}$, with $h_{t-1} = \sum_j \Pr(S_{t-1} = j \mid Y_{t-2}) h_{j,t-1}$ (common mean)

Prerequisites for Today (7/7): Gibbs sampling

- ▣ **Gibbs Albert and Chib (1993), Chib (1996)**: draw $S \mid \theta$ by forward filtering-backward sampling; rows of $P \sim \text{Dirichlet/Beta}(a + n_{ii}, b + n_{ij})$
 - ▶ θ : all regime parameters; n_{ij} : the number of $i \rightarrow j$ transitions in the drawn path; a, b : prior parameters
- ▣ Regime means: $\mu_j \mid S \sim N(m_j, v_j)$, $v_j^{-1} = v_0^{-1} + n_j/\sigma_j^2$, $m_j = v_j(m_0/v_0 + n_j\bar{y}_j/\sigma_j^2)$
 - ▶ m_0, v_0 : prior mean and variance; n_j : number of periods in regime j ; $\bar{y}_j$: their average; precisions add, means average
- ▣ Label switching in MCMC: the sampler may swap regimes; identify afterwards by an ordering **Frühwirth-Schnatter (2001)**

A1: the chain and the first filter step [Solved]

Task

- $y_t = \mu_{S_t} + \varepsilon_t$, $\varepsilon_t \sim N(0, 1)$, $\mu_1 = 1$ (expansion), $\mu_2 = -1$ (recession), $p_{11} = 0.9$, $p_{22} = 0.75$; $y_1 = 0.5$.
- 1. Compute the ergodic probabilities and the expected durations.
- 2. With $\hat{\xi}_{1|0} = \pi$, compute η_1 , $f(y_1)$ and $\hat{\xi}_{1|1}$.
- 3. Give the first log-likelihood contribution.
- Report: two probabilities and two durations; two densities, one density and two probabilities; one number.

Solution

- 1. $\pi_1 = 0.25/0.35 = 0.714$, $\pi_2 = 0.286$; durations $1/0.1 = 10$ and $1/0.25 = 4$
- 2. $\eta_1 = (\phi(-0.5), \phi(1.5)) = (0.3521, 0.1295)$;
 $f(y_1) = 0.714 \cdot 0.3521 + 0.286 \cdot 0.1295 = 0.2885$;
 $\hat{\xi}_{1|1} = (0.872, 0.128)$
- 3. $\ln f(y_1) = -1.243$
- Interpretation: a growth of 0.5 is closer to the expansion mean, so the expansion probability rises from the prior 0.714 to 0.872

A2: the second step and the Kim smoother [Proposed]

Solution discussed in class

Task

- ▣ The model of A1 with $y_2 = -1.5$. Model: A1.
- ▣ 1. Compute $\hat{\xi}_{2|1} = P' \hat{\xi}_{1|1}$ and $\hat{\xi}_{2|2}$.
- ▣ 2. Take $T = 2$ and compute the smoothed $\hat{\xi}_{1|2}$ with the Kim formula.
- ▣ 3. Compute the four joint probabilities $\Pr(S_1 = i, S_2 = j \mid Y_2)$ and check that they sum to 1.
- ▣ Report: two vectors, one vector, a 2×2 table and one sentence on why the smoothed expansion probability of period 1 fell.

A3: one EM step [Solved]

Task

- ▣ $T = 5$, $y = (1.2, 0.8, -0.9, -1.4, 0.6)$; smoothed $\hat{\xi}_{1,t|T} = (0.9, 0.8, 0.1, 0.05, 0.7)$; joint $\Pr(S_{t-1} = 1, S_t = 1 \mid Y_T) = (0.75, 0.08, 0.04, 0.05)$ for $t = 2, \dots, 5$.
- ▣ 1. Update p_{11} .
- ▣ 2. Update μ_1, μ_2 (MSIH without regressors).
- ▣ 3. Update σ_1^2, σ_2^2 .
- ▣ Report: one number, two numbers, two numbers.

Solution

- ▣ 1. $p_{11} = 0.920/1.850 = 0.497$ (expected $1 \rightarrow 1$ transitions over expected visits to 1 at $t = 1..4$)
- ▣ 2. $\mu_1 = \sum \hat{\xi}_{1t} y_t / \sum \hat{\xi}_{1t} = 0.776$, $\mu_2 = -0.686$ (weights $1 - \hat{\xi}_{1t}$)
- ▣ 3. $\sigma_1^2 = 0.275$, $\sigma_2^2 = 0.742$ (weighted mean squared deviations)
- ▣ Interpretation: each M-step is a set of weighted averages; with five points the estimate of p_{11} is very imprecise

A4: regime forecasts and persistence [Proposed]

Solution discussed in class

Task

- ▣ The chain and means of A1; today the economy is in recession ($S_t = 2$ known). Model: A1, A3.
- ▣ 1. Show that $\xi_t = 1\{S_t = 1\}$ follows $\xi_t = (1 - p_{22}) + \lambda\xi_{t-1} + v_t$ with $\mathbb{E}[v_t | S_{t-1}] = 0$; give λ .
- ▣ 2. Compute $\Pr(S_{t+h} = 1 | S_t = 2)$ and $\mathbb{E}[y_{t+h} | S_t = 2]$ for $h = 1, 4, 12$.
- ▣ 3. Give the limit as $h \rightarrow \infty$.
- ▣ Report: one derivation and one number; six numbers; one number.

A5: how many regimes? [Solved]

Task

- ▣ $T = 200$; $\ln L$ for $K = 1, 2, 3$: $-310.4, -296.1, -292.8$ with 3, 7, 13 parameters; a bootstrap of the LR for $K = 2$ against $K = 3$ gives 6 of 199 simulated values above the observed one.
- ▣ 1. Compute AIC, BIC and HQ for each K and the choice of each criterion.
- ▣ 2. Compute LR for 1 against 2 and for 2 against 3; say why χ_4^2 is not the right reference for the first.
- ▣ 3. Compute the bootstrap p -value.
- ▣ Report: a table of nine values and three choices; two numbers and one sentence; one number.

Solution

- ▣ 1. AIC: 626.8, 606.2, 611.6; BIC: 636.7, 629.3, 654.5; HQ: 630.8, 615.5, 629.0: all choose $K = 2$
- ▣ 2. LR = 28.6 and 6.6; under $K = 1$ the transition probabilities are not identified and the score is zero, so the 9.49 of χ_4^2 is not valid
- ▣ 3. $p = (1 + 6)/(199 + 1) = 0.035$: the bootstrap rejects 2 regimes at 5% although all criteria prefer 2

A6: identification and the expanded state [Proposed]

Solution discussed in class

Task

- ▣ Hamilton's $\text{MSM}(K)\text{-AR}(4)$ and the $\text{MSIAH}(K)\text{-AR}(4)$, $K = 2$ and $K = 3$. Model: A5.
- ▣ 1. How many equivalent likelihood modes does label switching create?
- ▣ 2. How many states does the filter of $\text{MSM-AR}(4)$ track, and why does $\text{MSIAH-AR}(4)$ need only K ?
- ▣ 3. Count the free parameters of both models.
- ▣ Report: two numbers; two numbers and one sentence; four numbers.



A7: MS-GARCH, one step by hand [Solved]

Task

- ▣ Two regimes, $(\omega, \alpha, \beta) = (0.02, 0.05, 0.9)$ and $(0.2, 0.1, 0.8)$; $\varepsilon_{t-1} = 1.5$; $h_{1,t-1} = 0.5$, $h_{2,t-1} = 2$; $\Pr(S_{t-1} = j \mid Y_{t-2}) = (0.7, 0.3)$.
- ▣ 1. Compute $h_{1,t}$, $h_{2,t}$ in the Haas–Mittnik–Paoletta model.
- ▣ 2. Compute the collapsed h_{t-1} and $h_{1,t}$, $h_{2,t}$ in Gray's model.
- ▣ 3. How many regime histories would the naive model need after 10 and after 100 periods?
- ▣ Report: two numbers; three numbers; two numbers.

Solution

- ▣ 1. $h_{1,t} = 0.02 + 0.05 \cdot 2.25 + 0.9 \cdot 0.5 = 0.5825$;
 $h_{2,t} = 0.2 + 0.1 \cdot 2.25 + 0.8 \cdot 2 = 2.0250$
- ▣ 2. $h_{t-1} = 0.7 \cdot 0.5 + 0.3 \cdot 2 = 0.95$; $h_{1,t} = 0.9875$,
 $h_{2,t} = 1.1850$
- ▣ 3. $2^{10} = 1024$, $2^{100} \approx 1.27 \times 10^{30}$
- ▣ Interpretation: Gray pulls both regime variances towards the mixture: the calm regime inherits turbulence and the turbulent one forgets it

A8: Gibbs full conditionals [Proposed]

Solution discussed in class

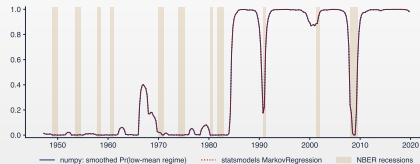
Task

- ▣ A draw of S has $n_{11} = 46$, $n_{12} = 4$ transitions out of regime 1, and $n_1 = 30$ observations in regime 1 with mean $\bar{y}_1 = 0.9$; prior $p_{11} \sim \text{Beta}(8, 2)$, $\mu_1 \sim N(0, 10)$, $\sigma_1^2 = 0.25$ fixed. Model: A3.
- ▣ 1. Give the full conditional of p_{11} , its mean, standard deviation and the implied duration.
- ▣ 2. Give the full conditional of μ_1 .
- ▣ 3. Explain why the prior on p_{11} also protects against the degenerate maxima of ML.
- ▣ Report: a distribution and three numbers; a distribution; one sentence.

B1: numpy against statsmodels [Solved]

- ▣ **Question:** do a transparent numpy implementation and statsmodels give the same answer for US GDP growth?
- ▣ **Inputs:** $100 \times \log$ change of GDPC1, 1947Q2–2019Q4 ($T = 291$); two regimes in mean and variance, no AR term
- ▣ **Tasks**
 1. Estimate the model by EM from 15 random starts, then polish by numerical ML.
 2. Estimate it with statsmodels MarkovRegression (20 random searches) and compare parameters and log-likelihoods.
 3. Compare the smoothed probabilities and compute the QPS against the NBER quarters.
 4. Interpretation: are the regimes recessions?
- ▣ **Report:** two parameter sets, three log-likelihoods, one difference, one QPS, one sentence
- ▣ **Notebook:** section B1

B1: solution [Solved]



Estimate	μ_1	μ_2	p_{11}	p_{22}	$\ln L$
numpy (EM + ML)	0.743	0.808	0.975	0.978	-348.50
statsmodels	0.743	0.808	0.975	0.978	-348.50

- EM alone: $\ln L = -348.57$ after 31 iterations; largest difference of the smoothed probabilities $< 10^{-4}$; $\sigma = 0.440$ and 1.196 ; QPS of the low-mean regime 1.096
- Interpretation: the two implementations agree
 - ▶ the regimes have almost the same mean and last 39.3 and 44.7 quarters
 - ▶ they are volatility eras (before and after the mid-1980s), not recessions; the QPS against NBER is poor

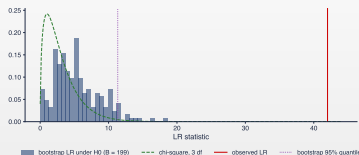
B2: Hamilton (1989) on his data and after 1984 [Proposed]

- **Question:** does Hamilton's model still find recessions in the Great Moderation?
- **Inputs:** Hamilton's real GNP data 1951Q2–1984Q4 (statsmodels); GDPC1 1984Q1–2019Q4; MSM(2)-AR(4); model: B1
- **Tasks**
 1. Estimate the model on Hamilton's data with the numpy expanded-state filter and with statsmodels MarkovAutoregression.
 2. Report the two regime means, the two persistence probabilities and the QPS against the NBER quarters.
 3. Re-estimate on 1984–2019 and report the same quantities and the share of quarters in the low regime.
 4. Interpretation: what does the low regime describe after 1984?
- **Report:** two log-likelihoods, two sets of five numbers, one share, one sentence
- **Notebook:** section B2

B3: how many regimes in Romanian GDP growth? [Solved]

- **Question:** is there statistical evidence for regimes in Romanian quarterly GDP growth?
- **Inputs:** Eurostat real GDP, seasonally and calendar adjusted, $100 \times \log$ change, 1995Q2–2026Q2 ($T = 125$); regimes in mean and variance
- **Tasks**
 1. Estimate $K = 1, 2, 3$ and compute AIC, BIC and HQ.
 2. Compute the LR statistic of $K = 1$ against $K = 2$ and its parametric bootstrap p -value ($B = 199$).
 3. Compare the bootstrap 95% quantile with the χ^2_3 critical value.
 4. Interpretation: which quarters form the low regime, and what does that say about the regimes?
- **Report:** a table of nine criteria, one LR and one p -value, two quantiles, a list of quarters, one sentence
- **Notebook:** section B3

B3: solution [Solved]



K	AIC	BIC	HQ
1	611.0	616.7	613.3
2	577.0	594.0	583.9
3	575.9	609.8	589.7

- LR = 42.01, bootstrap $p = 0.005$; bootstrap 95% quantile 11.34 against 7.81 for χ^2_3 ; regimes: $\mu = 0.03$ and 1.04, $\sigma = 3.92$ and 1.31, $p_{ij} = 0.90$ and 0.94
- Interpretation: the low regime contains 49 quarters
 - ▶ 1995Q2, 1995Q3, 1995Q4, 1996Q1, 1996Q2, 1996Q3, 1996Q4, 1997Q1, ...
 - ▶ the two regimes separate a volatile era with deep falls from stable growth, not recurring recessions

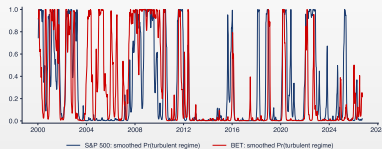
B4: Romanian inflation regimes [Proposed]

- ▣ **Question:** do the inflation regimes differ only in level and volatility, or also in persistence?
- ▣ **Inputs:** Eurostat HICP, annual inflation $100 \ln(P_t/P_{t-12})$, monthly from January 1997; three regimes; model: B3
- ▣ **Tasks**
 1. Estimate MSIH(3)-AR(1) (common AR coefficient) and MSIAH(3)-AR(1) (switching AR coefficient) by EM from 20 starts.
 2. Report the log-likelihoods, the BIC, the regime levels $\nu_j/(1 - \phi_j)$, the AR coefficients and the durations.
 3. Report the filtered regime probabilities at the end of the sample.
 4. Interpretation: which model would you use to forecast inflation in 2027, and why?
- ▣ **Report:** two log-likelihoods and BICs, two sets of levels, coefficients and durations, two probability vectors, one sentence
- ▣ **Notebook:** section B4

B5: bull and bear regimes of the S&P 500 and the BET [Solved]

- **Question:** are the turbulent regimes of the US and Romanian equity markets synchronised?
- **Inputs:** weekly log returns (Friday closes), 2000–2026, EODHD; two regimes in mean and variance, ordered by variance; model: B1
- **Tasks**
 1. Estimate the model for each index (EM from 15 starts and a numerical polish).
 2. Report annualised means and volatilities, expected durations and the share of turbulent weeks.
 3. Compute the correlation of the two turbulent probabilities and the maximum probability in 2008, 2020 and 2022.
 4. Interpretation: are BET regimes imported from the US?
- **Report:** a table of two rows and seven columns, one correlation, six probabilities, one sentence
- **Notebook:** section B5

B5: solution [Solved]



Index	μ calm	σ calm	μ turb.	σ turb.	durations (weeks)	% turb.	2008, 2020, 2022
S&P 500	16.6	10.9	-20.6	28.3	37; 14	26	1.00; 1.00; 1.00
BET	20.1	13.6	3.2	39.9	33; 12	25	1.00; 1.00; 1.00

- Correlation of the turbulent probabilities 0.48; both turbulent in 15% of weeks, BET alone in 11%
- Interpretation: the big global episodes are shared, but a large part of BET turbulence is domestic; regimes are mostly volatility regimes with imprecise means

B6: MS-GARCH for the BET [Proposed]

- **Question:** how much of the GARCH persistence of the BET is regime persistence?
- **Inputs:** BET daily log returns (%), 2010–2026, $T = 4186$; GARCH(1,1) and the two-regime GARCH of Haas, Mittnik and Paolella; model: B5, A7
- **Tasks**
 1. Estimate both models by numerical ML with Normal innovations.
 2. Report $\alpha + \beta$ for GARCH and in each regime, the persistence probabilities, the regime volatilities and the BIC.
 3. Report the share of days in the high-volatility regime.
 4. Interpretation: what does this imply for a 10-day volatility forecast after a calm month?
- **Report:** two BICs, three persistences, two probabilities, two volatilities, one share, one sentence
- **Notebook:** section B6

C1: EUR/RON in 2025, a new regime or a break? [Proposed]

- ▣ **Question:** was the depreciation of May 2025 the start of a new volatility regime, a one-off level shift, or both?
- ▣ **Inputs:** BNR reference rate, daily log changes (%), January 2023 – September 2026 ($T = 926$); model: B5
- ▣ **Tasks**
 1. Write a five-line pre-registration: hypotheses, models (MSIH(2), a level break), sample, criteria, what would change your mind.
 2. Estimate the two-regime model and report the regime volatilities, durations, the share of turbulent days before and after May 2025.
 3. Interpretation: what can and cannot be concluded from one episode?
- ▣ **Report:** a five-line plan, a table of six numbers, a project plan
- ▣ **Notebook:** section C1

C2: audit an AI answer [Proposed]

- ▣ A student asked an AI assistant to summarise Markov-switching models. The answer:
- ▣ (a) The smoothed probabilities show what a forecaster knew in real time.
- ▣ (b) With $p_{11} = 0.9$ and $p_{22} = 0.75$, the ergodic probability of regime 1 is 0.9.
- ▣ (c) To test one regime against two, compare the LR statistic with a chi-square with 4 degrees of freedom.
- ▣ (d) EM always reaches the global maximum because each step increases the likelihood.
- ▣ (e) Hamilton's MSM-AR(4) with two regimes needs a filter over 32 states.
- ▣ (f) Gray's MS-GARCH is exact; the model of Haas, Mittnik and Paoletta approximates the path.
- ▣ (g) A positive GPH estimate of d proves long memory rather than regime switching.
- ▣ Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
 - ▶ 2. Report: a list of seven verdicts with one line of justification each.

Key takeaways

- ▣ The Hamilton filter is Bayes' rule applied each period; the Kim smoother runs it backwards
- ▣ EM is a sequence of weighted averages; start it many times
- ▣ The number of regimes is not tested with a χ^2 count
- ▣ Regimes found by a model are statistical objects: read them against events before naming them
- ▣ An AI answer is a draft: check filtered against smoothed, ergodic against persistence, local against global

After the seminar

- ▣ Lecture 7 develops each topic of today: derivations, testing, TVTP, MS-VAR, MS-GARCH, Bayesian estimation, long memory, forecasting
- ▣ Try the [Proposed] tasks in the notebook
- ▣ C1 can grow into a team project: exchange-rate regimes of the leu, 1999–2026
- ▣ Reading: [Hamilton \(1989\)](#); [Hamilton \(1990\)](#); [Kim \(1994\)](#); [Haas, Mittnik and Paoletta \(2004\)](#)
- ▣ **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

References

- ▣ Albert, J. H., & Chib, S. (1993). Bayes inference via Gibbs sampling of autoregressive time series subject to Markov mean and variance shifts. *Journal of Business & Economic Statistics*, 11(1), 1–15. doi:10.1080/07350015.1993.10509929
- ▣ Chib, S. (1996). Calculating posterior distributions and modal estimates in Markov mixture models. *Journal of Econometrics*, 75(1), 79–97. doi:10.1016/0304-4076(95)01770-4
- ▣ Frühwirth-Schnatter, S. (2001). Markov chain Monte Carlo estimation of classical and dynamic switching and mixture models. *Journal of the American Statistical Association*, 96(453), 194–209. doi:10.1198/016214501750333063
- ▣ Garcia, R. (1998). Asymptotic null distribution of the likelihood ratio test in Markov switching models. *International Economic Review*, 39(3), 763–788. doi:10.2307/2527399
- ▣ Gray, S. F. (1996). Modeling the conditional distribution of interest rates as a regime-switching process. *Journal of Financial Economics*, 42(1), 27–62. doi:10.1016/0304-405X(96)00875-6
- ▣ Haas, M., Mittnik, S., & Paolella, M. S. (2004). A new approach to Markov-switching GARCH models. *Journal of Financial Econometrics*, 2(4), 493–530. doi:10.1093/jjfinec/nbh020
- ▣ Hamilton, J. D. (1990). Analysis of time series subject to changes in regime. *Journal of Econometrics*, 45(1–2), 39–70. doi:10.1016/0304-4076(90)90093-9
- ▣ Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2), 357–384. doi:10.2307/1912559
- ▣ Hansen, B. E. (1992). The likelihood ratio test under nonstandard conditions: Testing the Markov switching model of GNP. *Journal of Applied Econometrics*, 7(S1), S61–S82. doi:10.1002/jae.3950070506
- ▣ Kim, C.-J. (1994). Dynamic linear models with Markov-switching. *Journal of Econometrics*, 60(1–2), 1–22. doi:10.1016/0304-4076(94)90036-1
- ▣ Krolzig, H.-M. (1997). *Markov-Switching Vector Autoregressions: Modelling, Statistical Inference, and Application to Business Cycle Analysis*. Springer. doi:10.1007/978-3-642-51684-9